

Daniel W Liu

Computer Science and Finance Student

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EDUCATION

University of Waterloo <i>Double Major in Computer Science and Finance</i> <i>Bachelor of Computing and Financial Management (Co-op)</i> — GPA: 3.94/4.0	Sept 2025 – April 2029 Waterloo, ON
International Baccalaureate Diploma Programme <i>IB Diploma</i> — 45/45 Perfect Score — <i>Top 0.5% Internationally</i>	Sept 2021 – May 2025 Thornhill, ON

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, TypeScript, GLSL, SQL, HTML, CSS
Frameworks: React.js, Next.js, Node.js, Express.js, FastAPI, Tailwind CSS
Cloud and Databases: AWS, Google Cloud, PostgreSQL, MongoDB, Redis, Supabase, Drizzle
Data and Libraries: OpenGL, pandas, NumPy, OpenCV, Matplotlib, yFinance, PyQt5, Three.js
DevOps and Tools: Docker, Git, Linux, CI/CD, Jest, Jenkins, Vercel, GitHub, Unity, Blender

EXPERIENCE

Software Developer Intern - Builds, Infrastructure and Tooling <i>PlayStation (Sony Interactive Entertainment)</i>	May 2026 – Aug 2026 Waterloo, ON
Software Engineering Intern <i>Wedge (YC S25)</i> - Shipped a full-stack AI voice platform using Next.js, Supabase, and Vapi, meeting weekly with 3 hospital clients to auto-navigate insurance claim calls and extract 45+ structured claim data fields across 50+ clinical sites. - Reduced server polling overhead by 67% and resolved race conditions through atomic Postgres operations, while building an Interactive Voice Response engine that auto-maps call flows into 8-node AI decision trees.	Jan 2026 – Apr 2026 San Francisco, CA
Software Engineering Intern <i>HOSA Canada</i> - Engineered a scalable full-stack educational resource sharing platform using Next.js, AWS, and MongoDB, serving 9,000+ Canadian students competing across 50+ unique associations spanning North America and Asia. - Deployed a production MERN workshop sign-up system processing 19,943 registrations across 347 workshop sessions, integrating Twilio email notifications, Redis queues, PM2 cron workers, and 457 Jest unit tests.	Sept 2025 – Dec 2025 Toronto, ON
Machine Learning Research Intern <i>Pearl Lab Holland Bloorview Kids Rehabilitation Hospital</i> - Developed real-time AI object detection systems using OpenCV and pupil-apriltags for <i>Bootle Blast</i>, a gamified rehabilitation platform clinically tested at 3+ Canadian hospitals for children's motor skill rehabilitation. - Benchmarked ML object detection models using NumPy and pandas against YOLOv8 on a dataset of 250+ unique environments, improving mean average precision (mAP) by 46.8% in real-world scenarios.	Jul 2025 – Aug 2025 Toronto, ON

PROJECTS

rainbolt.ai <i>Next.js, Three.js, FastAPI, OpenAI CLIP, Pinecone, LangChain, Google Cloud</i> Devpost GitHub - Awarded Best UI Hack at UofT Hack the Valley X as frontend lead, building an AI geolocation teaching platform with Next.js, Three.js, Zustand, and Gemini for interactive 3D visualizations and geographic reasoning. - Architected a RAG pipeline with a dataset of 955,199+ images, integrating OpenAI CLIP, Pinecone, LangChain, and Mapillary API with a FastAPI backend to analyze street-level imagery and generate visual location reasoning.
Multithreaded Voxel Engine <i>C++20, OpenGL 4.1, GLFW, GLM, Dear ImGui, FastNoiseLite, CMake</i> GitHub - Built a multithreaded C++20/OpenGL voxel engine with procedural terrain, shadows, and water, using a greedy meshing algorithm to merge identical block faces and cut GPU triangles 27x per chunk, sustaining 145 FPS. - Engineered a chunk-streaming pipeline with 9 worker threads (940 chunks/sec), infinite world streaming with bounded memory, raycast-based block place/break, and RLE-compressed binary saves (144x compression).
VibeTrade <i>Next.js, FastAPI, Supabase, LangGraph, Alpaca, Finnhub, ElevenLabs</i> Devpost GitHub - Built an agentic AI trading dashboard that analyzes market data, news, and portfolio context to generate real-time actionable insights and explainable trading rationales, integrating 5 financial APIs over live WebSocket streams. - Implemented a customizable 3D AI voice agent with low-latency voice synthesis, background workers for market monitoring, and a FastAPI + LangGraph workflow enabling voice-triggered trade execution and reasoning.